

Template-Based Mass Spectrometry: Three-Dimensional Molds as Geometric Filters in Molecular Flow

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Abstract

We introduce template-based mass spectrometry, a framework treating spectral analysis as geometric filtering through three-dimensional molds. Each template encodes expected patterns—*isotope distributions, adduct masses, fragmentation signatures*—as geometric constraints that ions must satisfy for identification. Unlike database matching, templates operate as continuous filters, quantifying the degree to which observed spectra conform to theoretical predictions. The framework achieves 94.7% identification accuracy on complex metabolomics samples while providing probabilistic confidence scores. We demonstrate that template composition follows algebraic rules, enabling systematic construction of templates for novel compound classes.

1 Introduction

Mass spectrometry identification traditionally relies on spectral database matching: observed spectra are compared against reference libraries, with matches scored by similarity metrics. This approach requires prior measurement of standards and fails for compounds absent from databases.

Template-based analysis inverts this paradigm. Rather than matching against stored spectra, we define geometric constraints—*templates*—that encode the structural and physical rules governing spectral patterns. A template for phosphatidylcholine specifies: (1) headgroup mass of 184.0733 Da, (2) characteristic neutral losses of 59.0735 and 183.0660 Da, (3) isotope pattern following binomial distribution. Observed spectra satisfying these constraints identify as phosphatidylcholine regardless of fatty acid composition.

This approach transforms identification from discrete database lookup to continuous geometric filtering, providing confidence scores and enabling identification of novel compounds within characterized classes.

2 Mathematical Framework

2.1 Template Definition

Definition 1 (Spectral Template). *A template T is a tuple $(M, \mathcal{I}, \mathcal{F}, \mathcal{A}, \phi)$ where:*

- M : mass function mapping molecular formula to exact mass
- $\mathcal{I}$: isotope pattern specification
- $\mathcal{F}$: fragmentation rule set
- $\mathcal{A}$: adduct formation rules
- ϕ : tolerance function

Each component defines geometric constraints in mass-intensity space:

$$\mathcal{I}(m, k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (1)$$

$$\mathcal{F}(m) = \{m - \Delta m_i : i \in \text{losses}\} \quad (2)$$

$$\mathcal{A}(m) = \{m + a_j : j \in \text{adducts}\} \quad (3)$$

where n is the number of heavy-isotope-contributing atoms, p is the isotope probability, and Δm_i , a_j are neutral loss and adduct masses.

2.2 Template Matching Score

Given an observed spectrum $S = \{(m_i, I_i)\}$ and template T , the matching score is:

$$\text{Score}(S, T) = \prod_{c \in \text{constraints}(T)} f_c(S) \quad (4)$$

where $f_c(S) \in [0, 1]$ measures constraint satisfaction.

For isotope patterns:

$$f_{\mathcal{I}}(S) = 1 - \frac{1}{K} \sum_{k=0}^{K-1} \left| \frac{I_{M+k}}{I_M} - \mathcal{I}(M, k) \right| \quad (5)$$

For fragmentation:

$$f_{\mathcal{F}}(S) = \frac{|\{m \in S : m \in \mathcal{F}(M) \pm \phi\}|}{|\mathcal{F}(M)|} \quad (6)$$

2.3 Template Algebra

Templates compose algebraically, enabling construction of complex templates from simpler components.

Proposition 1 (Template Product). *For templates T_1 and T_2 , the product template $T_1 \otimes T_2$ satisfies:*

$$\text{Score}(S, T_1 \otimes T_2) = \text{Score}(S, T_1) \cdot \text{Score}(S, T_2) \quad (7)$$

This enables modular template construction:

$$T_{\text{PC}} = T_{\text{headgroup}} \otimes T_{\text{fatty acid}} \otimes T_{\text{isotope}} \quad (8)$$

Proposition 2 (Template Sum). *The sum template $T_1 \oplus T_2$ matches spectra satisfying either template:*

$$\text{Score}(S, T_1 \oplus T_2) = \max(\text{Score}(S, T_1), \text{Score}(S, T_2)) \quad (9)$$

Template sums handle compound class ambiguity when multiple structural interpretations are valid.

3 Three-Dimensional Mold Representation

3.1 Geometric Visualization

Templates define three-dimensional surfaces in $(m/z, \text{RT}, I)$ space. A compound satisfies a template when its spectral coordinates lie within the template surface, analogous to a key fitting a lock.

The template surface Σ_T is:

$$\Sigma_T = \{(m, t, I) : \text{Score}((m, I), T) \geq \theta\} \quad (10)$$

where θ is the acceptance threshold.

3.2 Mold Deformation

Template parameters can vary continuously, creating a family of molds:

$$T(\alpha) = T_0 + \alpha \cdot \delta T \quad (11)$$

This enables:

- Mass shift correction for instrument calibration
- Intensity scaling for matrix effects
- Retention time adjustment for column aging

3.3 Partition Coordinate Integration

Templates connect to partition coordinates through the mapping:

$$T \mapsto (n_T, \ell_T, m_T, s_T) \quad (12)$$

where template coordinates derive from:

$$n_T = \lfloor \sqrt{M_T/m_{\text{ref}}} \rfloor + 1 \quad (13)$$

$$\ell_T = |\mathcal{F}(M_T)| \bmod n_T \quad (14)$$

$$m_T = |\mathcal{I}(M_T)| - \ell_T \quad (15)$$

$$s_T = \text{sign}(z_T) \quad (16)$$

This integration enables S-entropy calculation for template-matched ions.

4 Ion Journey Visualization

4.1 Eight-Stage Pipeline

Template-based analysis tracks each ion through eight distinct processing stages:

Stage 1: Sample Injection. Ion enters as an uncharacterized entity with measured m/z and intensity.

Stage 2: Chromatographic Separation. Retention time assignment provides temporal coordinate, constraining possible identifications.

Stage 3: Ionization. Adduct and charge state determination through template matching against $\mathcal{A}$.

Stage 4: MS1 Analysis. Isotope pattern matching against $\mathcal{I}$ to confirm molecular formula.

Stage 5: S-Entropy Calculation. Computation of (S_k, S_t, S_e) coordinates from spectral features.

Stage 6: Partition Assignment. Mapping to categorical state (n, ℓ, m, s) in partition space.

Stage 7: Thermodynamic Validation. Verification of physical plausibility through dimensionless numbers (We, Re, Oh).

Stage 8: Droplet Transformation. Final representation as information droplet with geometric properties.

4.2 Stage-Specific Template Application

Each stage applies specific template components:

$$\text{Stage 3: } T_{\text{adduct}} \quad (17)$$

$$\text{Stage 4: } T_{\text{isotope}} \quad (18)$$

$$\text{Stage 5: } T_{\text{entropy}} \quad (19)$$

$$\text{Stage 7: } T_{\text{thermodynamic}} \quad (20)$$

The cumulative template score tracks identification confidence through the pipeline:

$$\text{Score}_{\text{cumulative}}(k) = \prod_{i=1}^k \text{Score}_i \quad (21)$$

5 Algorithm Implementation

5.1 Template Matching Algorithm

Algorithm 1 Template-Based Identification

Require: Spectrum S , Template library $\mathcal{T}$, Threshold θ

Ensure: Identification list with scores

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1: candidates  $\leftarrow \emptyset$ 
2: for each  $T \in \mathcal{T}$  do
3:    $s_{\text{iso}} \leftarrow f_{\mathcal{I}}(S, T)$ 
4:   if  $s_{\text{iso}} < \theta_{\text{iso}}$  then
5:     continue
6:   end if
7:    $s_{\text{frag}} \leftarrow f_{\mathcal{F}}(S, T)$ 
8:    $s_{\text{add}} \leftarrow f_{\mathcal{A}}(S, T)$ 
9:    $s_{\text{total}} \leftarrow s_{\text{iso}} \cdot s_{\text{frag}} \cdot s_{\text{add}}$ 
10:  if  $s_{\text{total}} \geq \theta$  then
11:    candidates  $\leftarrow$  candidates  $\cup \{(T, s_{\text{total}})\}$ 
12:  end if
13: end for
14: return sort(candidates, descending by score)

```

5.2 Computational Complexity

Template matching complexity is:

$$O(|S| \cdot |\mathcal{T}| \cdot \max_T |\text{constraints}(T)|) \quad (22)$$

For typical parameters ($|S| \approx 10^3$, $|\mathcal{T}| \approx 10^4$, constraints ≈ 10), this yields 10^8 operations—tractable for real-time analysis.

5.3 Template Indexing

Performance improves through mass-based indexing:

$$\mathcal{T}[m] = \{T \in \mathcal{T} : |M_T - m| \leq \phi\} \quad (23)$$

This reduces effective library size to $O(1)$ templates per query mass, yielding $O(|S|)$ complexity.

6 Experimental Validation

6.1 Benchmark Datasets

Validation used four datasets spanning different platforms and compound classes:

- HILIC negative mode metabolomics (Lab A)
- Waters qTOF phospholipidomics (Lab B)
- Thermo Orbitrap triglycerides (Lab C)
- BSA proteomics standard (Lab D)

6.2 Identification Accuracy

Table 1: Template-based identification performance

Dataset	Precision	Recall	F1
HILIC metabolomics	0.93	0.91	0.92
Phospholipids	0.96	0.94	0.95
Triglycerides	0.95	0.93	0.94
BSA proteomics	0.94	0.96	0.95
Overall	0.947	0.935	0.941

Template-based identification achieves 94.7% precision across diverse compound classes.

6.3 Platform Independence

Critical validation: templates developed on one platform transfer to others without modification.

Table 2: Cross-platform template transfer

Training $\rightarrow$ Test	Same-platform F1	Cross-platform F1
Waters $\rightarrow$ Waters	0.95	—
Waters $\rightarrow$ Thermo	—	0.91
Thermo $\rightarrow$ Waters	—	0.90

Cross-platform performance degrades by only 4–5%, demonstrating template transferability.

6.4 Novel Compound Identification

Template-based analysis identifies compounds absent from spectral databases. For 127 metabolites without HMDB spectral records:

- 94 (74%) correctly assigned compound class
- 67 (53%) correctly assigned subclass
- 41 (32%) correctly assigned specific structure

Class-level identification enables biological interpretation even without exact structural assignment.

7 Ion Journey Case Study

7.1 Single Ion Through Eight Stages

We trace ion m/z 760.5851, RT 14.2 min, from a phospholipid sample:

Stage 1: Injection as neutral phospholipid in methanol.

Stage 2: Elution at RT 14.2 min, consistent with PC class ($\ell = 3$).

Stage 3: Ionization as $[M+H]^+$, template match score 0.97.

Stage 4: Isotope pattern: $M + 1$ at 761.5884 (36.2% relative), $M + 2$ at 762.5918 (8.1% relative). Template score 0.94.

Stage 5: S-entropy coordinates: $S_k = 2.14$, $S_t = 0.87$, $S_e = 1.45$.

Stage 6: Partition assignment: ($n = 4, \ell = 3, m = 1, s = +1/2$).

Stage 7: Thermodynamic validation: $W_e = 847$, $Re = 1.2$, $Oh = 0.024$. All within physical bounds.

Stage 8: Droplet radius 4.7 nm, surface tension consistent with PC headgroup.

Cumulative template score: $0.97 \times 0.94 = 0.91$. Identification: PC 34:1.

7.2 Multi-Platform Comparison

The same phospholipid analyzed across three instruments shows:

- m/z variance: ± 2.3 ppm
- RT variance: ± 0.4 min (normalized)
- Template score variance: ± 0.03

Platform-independent identification confirms template robustness.

8 Template Library Construction

8.1 Lipid Templates

Lipid templates follow modular construction:

$$T_{\text{lipid}} = T_{\text{class}} \otimes T_{\text{FA chains}} \otimes T_{\text{backbone}} \quad (24)$$

For phosphatidylcholine:

$$T_{\text{PC}} = T_{184.0733} \otimes T_{\text{NL59}} \otimes T_{\text{glycerol}} \quad (25)$$

This generates 1,847 specific PC templates from three base components.

8.2 Metabolite Templates

Metabolite templates emphasize fragmentation patterns:

$$T_{\text{metabolite}} = T_{\text{formula}} \otimes T_{\text{substructure}} \otimes T_{\text{MS/MS}} \quad (26)$$

Common neutral losses define substructure templates:

$$T_{\text{H}_2\text{O}} : \Delta m = 18.0106 \quad (27)$$

$$T_{\text{CO}_2} : \Delta m = 43.9898 \quad (28)$$

$$T_{\text{NH}_3} : \Delta m = 17.0265 \quad (29)$$

8.3 Protein Templates

Peptide templates incorporate charge state series:

$$T_{\text{peptide}} = \bigotimes_{z=1}^{z_{\text{max}}} T_{[M+zH]^{z+}} \quad (30)$$

B/y ion series provide fragmentation templates:

$$T_{\text{MS/MS}} = \bigoplus_{i=1}^{n-1} (T_{b_i} \oplus T_{y_{n-i}}) \quad (31)$$

9 Discussion

9.1 Comparison to Machine Learning

Template-based analysis differs fundamentally from machine learning approaches:

Interpretability: Template scores decompose into physically meaningful components. ML models provide predictions without mechanistic insight.

Generalization: Templates encode rules, not patterns. Novel compounds satisfying rules are correctly identified. ML requires training examples.

Data efficiency: Template construction requires chemical knowledge, not large datasets.

9.2 Limitations

Template coverage: Compounds outside defined templates receive no score. Database matching can find unexpected similarities.

Isomer discrimination: Templates based on exact mass cannot distinguish structural isomers without MS/MS.

Expertise requirement: Template construction requires domain knowledge.

10 Conclusion

Template-based mass spectrometry provides a geometric framework for spectral interpretation. Three-dimensional molds encode expected patterns as continuous filters, quantifying identification confidence. Template algebra enables modular construction of complex templates from simple components. The framework achieves 94.7% identification accuracy while providing interpretable scores and enabling novel compound identification. Integration with partition coordinates and S-entropy creates a complete analytical pipeline from sample injection through categorical state assignment.

Data Availability

Template libraries and validation datasets are available at [repository URL].